
LECTURE NOTES

Discrete Mathematics notes

Induction

DM549 - Discrete Mathematics

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1 Induction (weak)

Proof technique used to establish that a statement $P(n)$ holds for all integers $n \geq m$. this involves 2 primary steps

- 1) Base case:
 - a) Prove that $P(m)$ holds
- 3) Inductive step:
 - a) Prove that $P(k) \rightarrow P(k + 1) \forall k \geq m$

1.1 Applying Induction

- 1) **Understand** the statement
 - a) Identify $P(n)$, the proposition you are trying to prove.
- 2) Prove the **Base case**
 - a) Substitute $n = m$ into $P(n)$
 - b) Verify that $P(m)$ is true.
- 3) State an **Inductive hypothesis**
 - a) Assume $P(k)$ holds for some arbitrary $k \geq m$. This is your inductive hypothesis.
- 4) Prove the **Inductive step**
 - a) Use the inductive hypothesis (**IH**) to show that $P(k) \implies P(k + 1)$
 - b) Explicitly demonstrate where $P(k)$ is used in your reasoning
- 5) **Conclude** the proof
 - a) State that $P(n)$ thereby holds for all $n \geq m$, based on the base case and the inductive step

1.2 Common pitfalls in weak induction proofs

- Overlooking conditions in the statement (e.g., $n \geq m$).
- Forgetting to verify the base case.
- Not clearly showing how $P(k)$ leads to $P(k + 1)$

Example: Proof by weak induction on sums

Proof by induction

Goal: Prove $P(n)$ that

$$\sum_n^{i=1} 2^i = 2^{n+1} - 1 \quad \forall n \geq 0$$

1 Base case

2 | $2^0 = 2^{0+1} - 1 = 1 \quad \top$

3 Inductive hypothesis

4 | Assume that $\sum_k^{i=1} 2^i = 2^{k+1} - 1$ for $k \geq 0$ (base case)

5 Inductive step

6 |
$$\begin{aligned} \sum_{k+1}^{i=0} 2^i &= \left(\sum_k^{i=0} 2^i \right) + 2^{k+1} && \text{(IH in parentheses)} \\ &= (2^{k+1} - 1) + 2^{k+1} \\ &= 2^{k+2} - 1 \end{aligned}$$

□

Since base case is true, and we have shown that for any $P(k) \implies P(k+1)$,

We have proven by induction that $\sum_n^{i=1} 2^i = 2^{n+1} - 1 \quad \forall n \geq 0$

2 Strong induction

Strong induction extends the weak induction by the fact that some inductive steps might need to look back more than 1 step (e.g. Fibonacci).

This follows two similar steps

Given $l \geq 0$ (how many extra steps back (beyond k) the inductive step needs.)

1) **Base case**

a) Prove that $P(i)$ holds for $m \leq i \leq m + l$

2) **Inductive step**

a) Prove that

$$\bigwedge_{i=k}^k P(i) \rightarrow P(k+1) \quad \forall k \geq m+l.$$

2.1 Common pitfalls in strong induction proofs

- Forgetting to verify all $l + 1$ base cases if multiple are needed.
- Failing to explicitly demonstrate where the strong inductive hypothesis is used in the proof of $P(k + 1)$
- Failing to incorporate all the needed $l + 1$ base cases into the proof of $P(n)$

2.2 Applying strong induction

1) **Understand** the statement

- a) Identify $P(n)$, the proposition you are trying to prove.
- b) If possible, determine the number of look-back steps l needed by the inductive step.

2) Prove the **Base case**

- a) Substitute $n = i$ into $P(n)$ for all i where $m \leq i \leq m + l$
- b) Verify that $P(m), P(m + 1), \dots, P(m + l)$ is true.

3) State a **Strong inductive hypothesis**

- a) Assume $P(i)$ holds $\forall i$ where $k - l \leq i \leq k$. This is your inductive hypothesis.

4) Prove the **Inductive step**

- a) Use the inductive hypothesis (**IH**) to show that

$$\bigwedge_{i=k-l}^k P(i) \implies P(k+1) \quad \forall k \quad \text{where } k \geq m+l$$

- b) Explicitly demonstrate where the strong inductive hypothesis is used in your reasoning.

5) **Conclude** the proof

- a) State that $P(n)$ thereby holds for all $n \geq m$, based on the base case and the inductive step.
- b) Demonstrate how the needed $l + 1$ base cases are used to validate $P(n)$

Example: Proof by strong induction on integers

Proof by induction

Prove that for every $n \in \mathbb{N}$ with $n \geq 4$, there exists $a, b \in \mathbb{N}$ such that $n = 2a + 5b$

Since if $n = 2a + 5b$, then $n + 2 = 2(a + 1) + 5b$, the inductive step increases n by 2.

To cover all integers ≥ 4 , we therefore need two consecutive base cases ($l = 1$).

1 **Base case**

2 | Verify for $n = 4$: Let $a = 2, b = 0$. Then $n = 2 \times 2 + 5 \times 0 = 4$ $\top$

3 | Verify for $n = 5$: Let $a = 0, b = 1$. Then $n = 0 \times 2 + 5 \times 1 = 5$ $\top$

4 **Strong inductive hypothesis**

5 | Assume that for all integers i with $k - 1 \leq i \leq k$ and $k \geq 5$, there exist $a, b \in \mathbb{N}$ such that $i = 2a + 5b$.

6 **Inductive step**

7 | By **IH** we have that $k - 1 = 2a + 5b$ for some $a, b \in \mathbb{N}$

8 | Then

9 |
$$k + 1 = (k - 1) + 2 = 2(a + 1) + 5b$$

10 | Since $a + 1 \in \mathbb{N}$ and $b \in \mathbb{N}$, $k + 1$ is in the required form. $\square$

By demonstrating that $k + 1$ can be expressed as $(k - 1) + 2 = 2(a + 1) + 5b$.

Thus using strong induction as verified $P(4)$ and $P(5)$, we have shown that $P(n)$ holds $\forall n \in \mathbb{N}$ with $n \geq 4$.

3 Structural induction

Structural induction is a proof technique specifically for recursively defined structured. These structures can be trees, lists or grammars etc. Much like regular kinds of inductions it involves two steps.

1) **Base case**

- a) Prove that $P(S_1)$ holds for the initial structure.

2) **Inductive step**

- a) Prove that $P(S_1) \implies P(S_{i+1})$ for all $i \geq 1$, using a structural inductive hypothesis

3.1 Common pitfalls in structural induction proofs

- Forgetting to verify the base case.
- Not demonstrating how the structural inductive hypothesis is applied
- Failing to cover all the recursive rules (if e.g. there are several rules)

3.2 Applying structural induction

1) **Understand** the *recursive* definition of the structure

- a) Identify the recursive rules defining the structure
b) State the property $P(s)$ to prove for all structures S .

2) Prove the **Base case**

- a) Verify that $P(S)$ holds for the initial structure.

3) State an **Structural inductive hypothesis**

- a) Assume $P(S_i)$ holds for all $i \geq 1$.

4) Prove the **Inductive step**

- a) Use the inductive hypothesis (**IH**) to prove that $P(S_{i+1})$ holds for the derived structure S_{i+1}
b) Explicitly demonstrate where the structural inductive hypothesis is used in your reasoning.

5) **Conclude** the proof

- a) State that $P(S_i)$ holds for all structures defined by recursive ruled based on the verified basis step and the structural inductive step.

Example: Proof by structural induction on a recursive set

Given the set $3 \in S$, which is defined recursively as follows

$$x, y \in S \implies x + y \in S$$

This defines a recursive $S = \{3, 6, 9, 12, 15, \dots\}$

Prove that this recursion is the same as $S' = \{3n \mid n \in \mathbb{Z}^+\}$

This is true iff $S \subseteq S'$ and $S' \subseteq S$

So we can define

$$P(k) = Q(k) \wedge Q'(k)$$

Where

$$Q(k) = (S_k \subseteq S') \text{ and } Q'(k) = (S'_k \subseteq S)$$

Proof by induction

Prove that $P(k) \forall i \in \mathbb{Z}^+$

1 Base case

2 For $P(1)$ where $S_1 = \{3\}$ and $S' = \{3\}$ we can simplify since $S_1 = S'_1$

3
$$P(1) = Q(1) \wedge Q'(1) = \underbrace{(S_1 \subseteq S'_1) \wedge (S'_1 \subseteq S_1)}_{\text{trivial since } \{3\}=\{3\}}$$

4 This is true.

5 Inductive hypothesis

6 Assume for some $k \geq 1$ that

7 $P(k)$ is true

8 This means that

9 $Q(k) \wedge Q'(k)$ is both true

10 Inductive step

11 Part 1

12 Given any $z \in S_{k+1}$ and the recursive definition

13
$$S_{k+1} = S_k \cup \{x + y \mid x, y \in S_k\}$$

14 We then have two cases:

15 Case 1 $Q(k)$

16 Let $z \in S_k$, then by **IH** since

17
$$S_k \subseteq S'$$

18
$$z \in S_k \implies z \in S'$$

19			Case 2 $Q'(k)$
20			Let any $z = x + y$ where $x, y \in S_k$
21			For x and y to exist there must be $a, b \in \mathbb{Z}^+$ such that
22			$z = 3a + 3b = 3(a + b)$
23			Since $a, b \in \mathbb{Z}^+$ then by IH , $z \in P' \implies S_{k+1} \subseteq S'$
24			Part 2
25			By definition $S'_{k+1} = \{3(k+1)\}$
26			Since $3(k+1) = 3k + 3$ and
27			$3k \in S \text{ and } 3 \in S$
28			Then
			$3(k+1) \in S \implies S'_{k+1} \subseteq S$

Thus by the basis step and the structural induction, it is proven that for any $k \geq 0$ it holds that $S' \subseteq S$ and $S \subseteq S'$